Smaller Wireless Power Transfer for Mobile Devices

Sponsor: student self-direct, no sponsor

Background:
As mobile phones continue to improve the size of these devices have greatly increased and the available space has become more and more valuable. This project seeks to provide an example of a place where space could be saved by demonstrating wireless power transfer on several different receiver networks.

Project Needs:
1. Design a wireless power transfer coil and inverter to function at up to 100kHz.
2. Develop and implement a control scheme using a microcontroller (or other pwm generation device) for the frequencies of 10kHz, 50kHz, and 100kHz.
3. Develop and implement switches and capacitor matching networks to allow for wireless power transfer at 10kHz, 50kHz, and 100kHz.
4. Design receivers and matching networks for the frequencies of 10kHz, 50kHz, and 100kHz to display.
5. Collect data on size of coil, power transferred, concern, etc.

Project Scope:
1. Inverter:
    - Create inverter that can operate at the frequencies of 10kHz, 50kHz, and 100kHz
    - Create matching networks & control schemes for the 3 frequencies
    - Create a containment for the device with switches to change between the frequencies
2. Transmitter:
    - Design a wireless power transfer transmitter coil
3. Receivers:
    - Design 3 wireless power transfer receiver coils (and matching networks) for the frequencies of 10kHz, 50kHz, and 100kHz
    - Create a rectifier circuit with a load that allows easy measurement of power

Potential Technical Considerations:
1. Inverter:
    - Power electronics design concerns (parasitic inductance, inverter design, filter design)
    - KiCad/Altium PCB design
2. Transmitter:
    - Coil design
    - LCR calculations of transmitter
3. Receivers:
    - KiCad/Altium PCB design
    - Choosing components that are “fair” for each frequency

Deliverables:

1. A wireless power transfer system in a containment with 3 switches to alternate between the three frequencies.
2. 3 receivers designed to receive the associated frequency.
3. Results and data from the testing of the 3 receivers on the wireless power transfer system.

Assumptions and Dependencies:
TBD